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ORIGINAL INVESTIGATION

Effect of Weekly Training Frequency With the Nordic Hamstring Exercise on Muscle-Strain Risk Factors in Football Players: A Randomized Trial

Thales M. Medeiros, João B. Ribeiro-Alvares, Carolina G. Fritsch, Gabriel S. Oliveira, Lucas Severo-Silveira, Evangelos Pappas, and Bruno M. Baroni

Purpose: To examine the differences between performing Nordic hamstring exercises once or twice a week on hamstring eccentric strength and other muscle-strain risk factors in high-level football players. ***Methods:*** In this randomized trial, 32 football players (18–23 y old) completed an 8-week Nordic hamstring exercise training program in 1 of 2 experimental groups: group 1 (once a week; n = 15) and group 2 (twice a week; n = 17). Knee-flexor/extensor peak torques and biceps femoris long-head muscle architecture were assessed through isokinetic dynamometry and ultrasonography, respectively, before and after the training programs. Analysis of covariance, effect sizes (ESs), and *t* tests for percentage change were used to assess the effect of the 2 interventions on the outcome measures. ***Results:*** Group 2 demonstrated higher hamstring concentric peak torque than group 1 posttraining (155–164 vs 149–158 N·m; $P = .043$; ES = 0.27), although there was also a statistical trend for higher hamstring eccentric peak torque (212–234 vs 198–221 N·m; $P = .098$; ES = 0.37), hamstring-to-quadriceps conventional ratio (0.56–0.59 vs 0.54–0.57; $P = .089$; ES = 0.31), and hamstring-to-quadriceps functional ratio (0.76–0.84 vs 0.71–0.79; $P = .076$; ES = 0.50). No between-groups differences were found for muscle thickness ($P = .864$; ES = 0.12), pennation angle ($P = .289$; ES = 0.18), fascicle length ($P = .406$; ES = 0.03), and quadriceps concentric peak torque ($P = .340$; ES = 0.02). ***Conclusion:*** Only the Nordic hamstring exercise training program performed twice a week strengthened the hamstrings of high-level football players, while similar changes in muscle architecture occurred with both once- and twice-weekly sessions.

Keywords: soccer, eccentric training, injury prevention

Hamstring strain injury (HSI) is the noncontact injury with the highest incidence in football (soccer).[1] A professional team with a 25-player squad typically suffers about 5 to 6 HSIs per season.[2] Even though there is increased awareness about these injuries in recent years, the HSI rates have actually increased.[3] On average, each injury keeps the athlete out of sports for around 17 days.[3] HSI also has high recurrence rates (~13% within 2 mo of return to play),[3] and the rehabilitation process is usually more complex for reinjured players.[4] Besides the direct impact on individual players' health and career, HSIs also negatively affect a team's performance[5] and finances.[6]

The biceps femoris long head (BF_{LH}) is the most commonly injured of the hamstring muscles.[7] Injuries typically occur during high-speed running, more specifically during the terminal swing phase when the hamstrings are required to contract forcefully while lengthening to decelerate the extending knee and flexing hip.[4] As a typical noncontact injury, there is a great interest in intrinsic factors that predispose athletes to HSI. Age and previous muscle injuries are well documented nonmodifiable risk factors,[8] while prospective studies have demonstrated that players with eccentric hamstring weakness,[9–12] muscle strength imbalance (hamstring to quadriceps [H:Q] ratio),[12–14] and reduced BF_{LH} fascicle length[11] are also more prone to sustaining HSIs.

Hamstring strain injuries are potentially preventable, and a recent meta-analysis by Al Attar et al[15] suggests that football teams including the Nordic hamstring exercise (NHE) could reduce HSI rates by up to 51%. This protective effect has been attributed to increased hamstring eccentric strength,[16–20] improved H:Q ratio,[16,18] and enhanced BF_{LH} fascicle length[16,19,21] generated by the NHE training programs. However, the response of both muscle strength and architecture to training is likely training status dependent; thus, extrapolating the research findings obtained in different populations to highly trained football players requires caution.[22] Therefore, randomized trials performed with high-level athletes have greater validity for the elite sport environment.

Findings from Bahr et al[23] suggest that only few top-level football teams use the NHE program whose effectiveness has been proven by the literature. In real practice, football clubs use multiple types of exercises and variations in terms of volume and intensity, and they adapt their programming as a function of several factors (eg, match schedule).[22] The evidence-based NHE programs comprise 10 weeks with 3 weekly sessions,[24,25] but premier division football teams rarely have a preseason longer than 4 to 6 weeks,[26,27] whereas in-season congested periods make it difficult to apply the NHE for 3 times a week.[23] Therefore, pragmatic programs of lower training weekly frequency are more feasible and have the potential for better compliance.

Nordic hamstring exercise training programs with 1 or 2 weekly sessions generate favorable muscular adaptations for HSI prevention in recreationally active adults,[16,28,29] amateur athletes,[19] and competitive young athletes.[30,31] Presland et al[29] reported that a low training volume (once a week) with the NHE is as effective as a high training volume (twice a week) for improving hamstring muscle strength and architecture, whereas Lacome et al[31] found significant effects of a training program with 1 weekly session regardless of the volume within exercise session. Minimizing the HSI risk through a

Medeiros, Ribeiro-Alvares, Fritsch, Oliveira, Severo-Silveira, and Baroni are with the Federal University of Health Sciences of Porto Alegre (UFCSPA), Porto Alegre, Brazil. Pappas is with the University of Sydney, Sydney, NSW, Australia. Baroni (bmbaroni@yahoo.com.br) is corresponding author.

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low-volume and pragmatic training program may represent a major breakthrough in the field of injury prevention in high-performance sport, and further investigation with high-level athletes is necessary. Therefore, this study aimed to examine the differences between performing NHE once or twice a week on hamstring eccentric strength and other muscle strain risk factors in high-level football players.

Methods

Study Design

A randomized trial was designed to determine the effects of NHE training programs performed once or twice a week on HSI risk factors in male football players. Volunteers allocated in group 1 (G1) performed 1 NHE training session per week, whereas volunteers in group 2 (G2) performed 2 weekly sessions. A control group (without NHE training) was not included in this study, because the club required that all players should be engaged in a preventive program using the NHE based on current evidence. However, previous evidence supports that outcomes included in the current study remain unchanged in a control group.[16]

Within 1 week before the start and 1 week after the end of the training programs, knee flexor/extensor strength and BF_{LH} muscle architecture of the left limb were assessed through isokinetic dynamometry and ultrasonography, respectively. No matches or resistance training sessions were performed in the 48-hour period before assessments. The NHE sessions were performed in the team's training field, and the evaluations were conducted at the physiotherapy laboratory of the Federal University of Health Sciences of Porto Alegre (Brazil). The study was approved by the institutional ethics committee.

Participants

A convenience sample of volunteers participated of this study, which was possible due to a partnership between our research group and a national premier division football team. Players engaged in the under-20 team (18–20 y old) and in the under-23 team (18–23 y old, considered a "transition team" for the main squad) were recruited. They all had professional contracts with the club and experience in national and international competitions. Both teams followed a similar training routine: 5 to 7 weekly practices on the field added to 2 or 3 strength and conditioning sessions in the gym, depending on the periodization phase and games calendar. The 8-week NHE training program comprised the preseason period (first 4 wk) and the first 4 weeks of the competitive season for both teams.

The inclusion criterion was that players should be participating regularly in the team's training routine during the data collection period. The following exclusion criteria were applied: (1) history of HSI within 6 months before the beginning of the study, (2) current musculoskeletal injury or illness that could impair the execution of NHE or isokinetic testing, and (3) noncompliance with the NHE training program (minimum of 7 training sessions for G1 and 14 training sessions for G2).

Fifty-four football players (23 from under-20 team and 31 from under-23 team) were able to participate. All of them agreed to participate through a signed consent form before data collection.

Sample-Size Estimation

Sample-size estimation was a priori completed using the primary outcome (ie, hamstring eccentric peak torque) and data from a previous study.[16] Power was set at 80% with an alpha level of .05, returning a minimum sample size of 10 subjects per group for a medium effect size (ES).

Randomization and Blinding

Participants were randomized into 2 experimental groups through the random.org website by an assistant not involved with assessment or training. Researchers involved in training sessions did not attend assessments and assessors did not attend the training sessions (ie, assessors were blinded to group allocation).

Outcomes

Hamstring eccentric peak torque was the primary outcome of the study. The secondary outcomes were hamstring and quadriceps concentric peak torques; H:Q conventional and functional ratios; and BF_{LH} muscle thickness, pennation angle, and fascicle length.

Assessments

As agreed with the coaching staff, players did not perform high-intensity practices the day before evaluations. Players were already familiarized with tests, and they were instructed not to take any medication (eg, analgesics) or consume alcoholic beverages within 24 hours before the tests or any stimulant substances (eg, coffee) on the day of the tests.

Muscle Architecture. A standardized protocol was used for ultrasound assessment by an experienced assessor,[32] with high test–retest reliability scores (intraclass correlation = .94–.98) for the BF_{LH} muscle architecture outcomes. A B-mode ultrasonography system (Vivid-i; GE Medical Systems, Fairfield, CT) with a linear array probe (frequency: 13 MHz; depth: 6 cm; width: 4 cm; GE 8L; GE Medical Systems) was used to assess muscle architecture parameters of BF_{LH}. After a minimum of 10 resting minutes, subjects were assessed in the prone position with the hips at neutral position, the knees fully extended, and the muscles relaxed. BF_{LH} length was determined as the distance between the ischial tuberosity and the superior border of the fibular head, and the ultrasound scans were taken at 50% of the muscle length. The ultrasound probe was positioned perpendicularly to the skin in the muscle's longitudinal axis, with a slight adjustment in probe orientation as necessary to optimize the aponeuroses and fascicle identification.

Three ultrasound images of the left thigh were taken and stored for analysis through the ImageJ software (National Institutes of Health, Bethesda, MD). Muscle thickness, pennation angle, and fascicle length from each image were assessed, and the average value of the 3 images was used for statistical analyses. Muscle thickness was calculated as the mean length of 3 mutually parallel lines perpendicular to the superficial and deep aponeurosis. The most visible fascicle in each image was used for measurement of fascicle pennation angle, and estimation of fascicle length via a validated equation.[33]

Muscle Strength. A standardized isokinetic testing protocol was used for strength assessment.[32] After a general warm-up (5 min of cycle-ergometer exercise), players were positioned seated on the isokinetic dynamometer (Biodex System 4; Biodex Medical Systems, Shirley, NY) according to the manufacturer's recommendations. Volunteers performed 10 concentric knee flexion/extension repetitions at 90 deg/s with a submaximal effort level

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for specific warm-up and familiarization with the equipment. Thereafter, 2 attempts of 3 consecutive maximum contractions were executed in the concentric–concentric mode (60°/s; 0°–90° of knee flexion) and 2 attempts of 3 consecutive maximal contractions in the eccentric–eccentric mode (60°/s; 30°–90° of knee flexion). Attempts were separated by 1-minute resting periods. Assessments were performed at 60°/s because slow angular velocities are recommended to assess the isokinetic peak torque (ie, the primary outcome of our study), and football players are commonly evaluated at this angular velocity.[34]

Quadriceps and hamstring peak torques were used for data analysis. H:Q conventional ratio (hamstring concentric peak torque/quadriceps concentric peak torque) and H:Q functional ratio (hamstring eccentric peak torque/quadriceps concentric peak torque) were calculated.[34]

Interventions

All players had been previously exposed to the NHE in the prior season. Even so, before the start of the training program, instruction on the NHE and a demonstration of its correct execution were given. Players were encouraged to slowly lower their torso to the floor level in a constant cadence (~3 s per repetition), sustaining a rigid torso position and only moving the knee joint while arms remained crossed on the chest (Figure 1). During the return to the NHE starting position, subjects used their upper limbs to avoid concentric actions of their hamstrings.

All players added the respective NHE training program to the regular training routine of their teams. Along the intervention period, players did not perform hamstring-specific exercises at the gym (eg, leg curl, stiff dead lift) to avoid interference on the outcomes. NHE training periodization was the same used by Severo-Silveira et al.[35] A similar training volume per NHE session (number of sets and repetitions) throughout the 8 weeks of training was followed by both groups (Figure 1). The only difference was that players in G1 performed a single NHE training session per week, whereas players in G2 trained twice a week. The days of the week intended for the NHE training sessions could vary over the training program due to the games calendar. However, a 72-hour interval period was always respected between the 2 weekly sessions in G2. NHE was always performed during warm-up (ie, before specific training). A researcher ensured the correct execution of the exercises and controlled the number of sets and repetitions performed by players.

Statistical Analysis

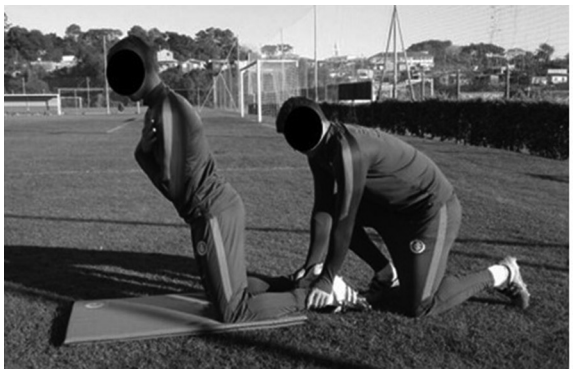
SPSS Statistics for Windows (version 20.0; IBM Corp, Armonk, NY) was used for the statistical analysis. After confirming that the assumptions were met, analysis of covariance tests were performed with baseline values as the covariates. Between-group comparisons were further assessed by percentage changes observed in each group ($\Delta\% = \text{posttraining/pretraining} - 1$) with independent *t* tests. Statistical significance was set at $P < .05$, whereas statistical trends were defined as *P* values between .05 and .10. Finally, we present ES calculations; training effects were classified as trivial (ES < 0.2), small (ES > 0.2), moderate (ES > 0.5), or large (ES > 0.8).[36]

Results

Fifteen players sustained injuries throughout the study period and were excluded. No injury occurred during the NHE training. Only 1 player (allocated into G1) sustained a HSI (overstretching mechanism during a match). In addition, 5 players were involved in transfers to other clubs and 2 players did not attend the posttraining evaluation due to personal reasons. Therefore, 32 players completed the study schedule: 15 in G1 and 17 in G2 (see flowchart in Figure 2). All these players had 100% attendance: 8 NHE training sessions for G1 and 16 training sessions for G2. Two players of G2 were excluded from muscle architecture analysis due to technical problems with their ultrasound images. Isokinetic outcomes had no missing data.

G1 and G2 were similar for age (18.80 [1.74] and 18.47 [1.07] y), body mass (78.80 [8.39] and 75.53 [10.44] kg), and height (1.82 [0.08] and 1.79 [0.10] m). Within-group analysis (Table 1) demonstrated that G2 increased the hamstring concentric and eccentric peak torques (small and moderate ESs, respectively) from pretraining to posttraining measurements, as well as H:Q conventional and functional ratios (small and moderate ESs, respectively). None of the isokinetic measures changed within the G1 (trivial ESs). NHE training increased BF_{LH} muscle thickness (small to moderate ESs) and fascicle length (small ESs) in both groups, whereas the BF_{LH} pennation angle did not change in either group (trivial ESs).

According to the analysis of covariance (Table 1), G2 demonstrated higher hamstring concentric peak torque than G1 posttraining ($P = .043$; small ES), while there was also a statistical trend for higher hamstring eccentric peak torque ($P = .098$; small ES), H:Q conventional ratio ($P = .089$; small ES), and H:Q functional ratio



Week	Frequency	Sets	Repetitions
1	1 or 2 *	2	6
2	1 or 2 *	3	6
3	1 or 2 *	3	8
4	1 or 2 *	3	10
5	1 or 2 *	4	8–10
6	1 or 2 *	4	8–10
7	1 or 2 *	4	10
8	1 or 2 *	4	10

Figure 1 — Players performing the Nordic hamstring exercise (left) and periodization of the Nordic hamstring exercise training program (right). *Frequency: once a week for group 1 and twice a week for group 2.

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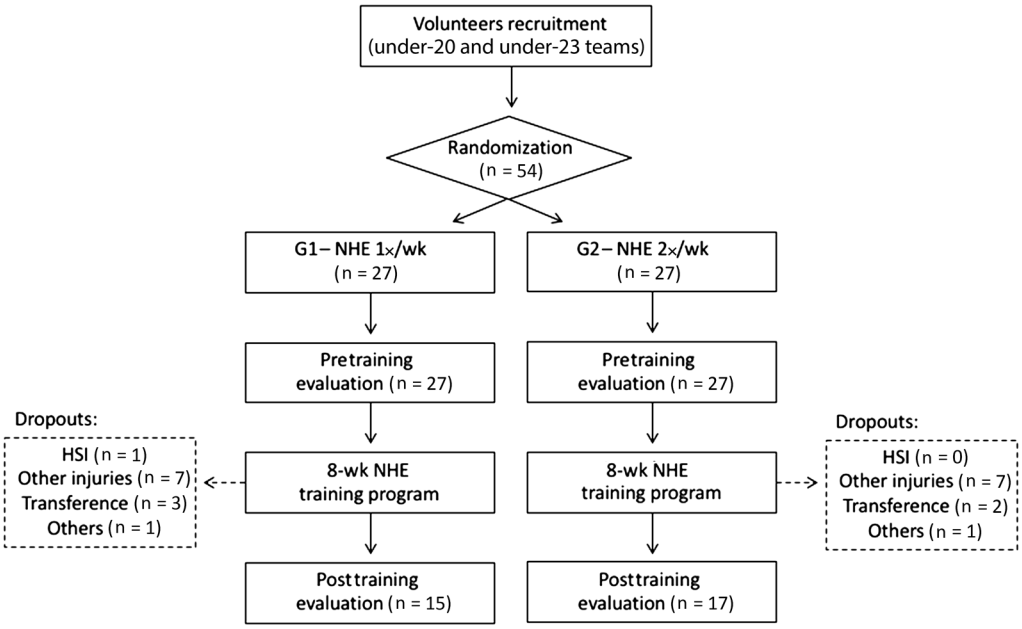

Figure 2 — Flowchart of the study. G1 indicates group 1; G2, group 2; HSI, hamstring-strain injury; NHE, Nordic hamstring exercise.

($P = .076$; small ES). No between-group differences were found for BF_{LH} muscle thickness ($P = .864$), pennation angle ($P = .289$), fascicle length ($P = .406$), and quadriceps concentric peak torque ($P = .340$), with trivial ES.

As illustrated in Figure 3, G2 had higher percentage increases than G1 for hamstring eccentric and concentric peak torques, as well as H:Q functional ratio ($P \leq .028$; large ESs). A statistical trend for higher H:Q conventional ratio in favor to G2 was also found ($P = .057$; moderate ES). There were no between-group differences for BF_{LH} muscle architecture outcomes and quadriceps concentric peak torque ($P \geq .699$; trivial ESs).

Discussion

This study aimed to compare the effects of 1-weekly session versus 2-weekly session NHE training programs on risk factors for HSI in high-level football players. Our main findings were that even though knee flexor strength and H:Q ratios improved only when the NHE was performed twice a week, both interventions resulted in BF_{LH} fascicle length and thickness increases.

The incidence of HSIs has increased among European top-level football teams.[3] The number of games and practice sessions per season have not changed since the beginning of the century,[3] thus the total time of players' exposure remains constant over the years. On the other hand, the physical demands in a competitive football match have increased. For instance, the high-intensity running and sprinting distances covered during a match increased by 30% to 35% between 2006–2007 and 2012–2013 seasons.[37] Considering that 70% of HSIs occurred during high-speed running in football,[7] it seems plausible that the increased HSIs rates may be related to the high intensity of modern football. It highlights the need for good conditioning and that coaching/medical staff should incorporate effective injury prevention programs supported by scientific evidence to counteract the injury risk factors.

Bahr et al[23] showed that the majority of premier division football teams are not fully compliant to an evidence-based hamstring injury prevention program of 3 weekly sessions for 10 weeks, followed by a weekly maintenance period during the rest of season.[24,25] The efficacy of the NHE and other eccentric exercises for prevention of HSIs depend on intervention compliance.[38] Therefore, the medical/coaching staff should be concerned not only with utilizing the NHE, but also with employing adequate training periodization to maximizing its benefits. However, the NHE training employed by competitive teams along the season may be influenced by several factors.[22] Match congestion periods, travels, coach compliance, and even the team's performance in competition sometimes have an impact on optimal periodization schedule. Thus, this study has focused on applying a more feasible program to the club's routine to make our findings applicable to the real-world of competitive football.

Insufficient eccentric strength has been traditionally suggested as an injury risk factor.[34] Findings from prospective studies in football,[11,12] Australian football,[10] and rugby[9] support that poor eccentric strength at preseason predisposes athletes to HSI during the season. In contrast, the role of H:Q ratio as a risk factor for HSI has been debated over the years. Although a prospective study reported no association between H:Q strength ratio and HSI,[39] there is evidence that athletes with agonist–antagonist strength imbalances are more prone to sustain a muscle injury.[12–14] Even in the face of conflicting evidence, medical staff from professional football clubs consider strength imbalance in the top 3 most important risk factors for noncontact injuries.[26,27]

In our study, hamstring isokinetic strength and H:Q ratios were improved only for players following the 2-weekly training program. Their hamstring eccentric strength improved ~11%, which is similar to the 11% to 13% increases observed following a 4- to 13-week NHE training in physically active volunteers[16,20] and professional football players.[17,18] In the current study, quadriceps strength was not altered in either group, whereas hamstring concentric and eccentric strength increased only in G2, resulting in improvements in both the H:Q conventional and functional ratios of this group.

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Table 1 Training Effects on Muscle Architecture and Strength Outcomes

	Group 1: NHE once a week				Group 2: NHE twice a week				
	Pre	Post	Within-group ES	Covariate adjusted 95% CI at posttraining	Pre	Post	Within-group ES	Covariate adjusted 95% CI at posttraining	Between-groups ES at posttraining (covariate adjusted)
Muscle strength									
H eccentric PT,[a] N·m	216.29 (31.35); 197–236	217.86 (38.24); 199–237	0.05	198–221	197.48 (40.81); 179–216	216.39 (34.17); 198–234	0.52[b]	212–234	0.37[c]
H concentric PT,[d] N·m	156.15 (21.71); 144–169	157.86 (20.46); 146–170	0.08	149–158	146.79 (25.73); 135–159	155.55 (25.73); 144–167	0.35[c]	155–164	0.27[c]
Q concentric PT, N·m	287.57 (47.29); 264–311	287.47 (43.53); 264–310	0.00	272–288	272.62 (42.84); 250–295	274.96 (43.72); 253–297	0.06	274–289	0.02
H:Q conventional ratio[a]	0.55 (0.07); 0.51–0.59	0.55 (0.07); 0.52–0.59	0.00	0.54–0.57	0.54 (0.08); 0.51–0.58	0.57 (0.07); 0.54–0.60	0.41[c]	0.56–0.59	0.31[c]
H:Q functional ratio[a]	0.76 (0.10); 0.70–0.82	0.76 (0.12); 0.71–0.82	0.00	0.71–0.79	0.73 (0.13); 0.67–0.79	0.79 (0.08); 0.74–0.84	0.57[b]	0.76–0.84	0.50[c]
Muscle architecture									
BF_{LH} thickness, cm	2.38 (0.24); 2.25–2.52	2.49 (0.16); 2.38–2.59	0.56[b]	2.39–2.53	2.31 (0.27); 2.17–2.45	2.42 (0.24); 2.31–2.52	0.45[c]	2.37–2.51	0.12
BF_{LH} pennation angle, deg	16.29 (3.14); 14.41–18.16	16.11 (4.27); 14.08–18.13	0.05	14.91–16.98	15.94 (3.91); 14.06–17.81	15.48 (3.32); 13.45–17.50	0.13	14.60–16.67	0.18
BF_{LH} fascicle length, cm	10.02 (1.84); 8.97–11.07	10.93 (2.92); 9.43–12.43	0.39[c]	10.26–11.68	10.09 (2.12); 9.03–11.14	11.04 (2.74); 9.54–12.54	0.40[c]	10.29–11.71	0.03

Abbreviations: BF_{LH}, biceps femoris long head; CI, confidence interval; ES, effect size; H, hamstrings; NHE, Nordic hamstring exercise; PT, peak torque; Q, quadriceps. Note: The values are presented as mean (SD); 95% CI.

[a] Trend for between-groups difference at post measurement after adjusting for baseline values. [b] Moderate effect size (ES > 0.05). [c] Small effect size (ES > 0.2). [d] Significant between-groups difference at post measurement after adjusting for baseline values ($P < .05$).

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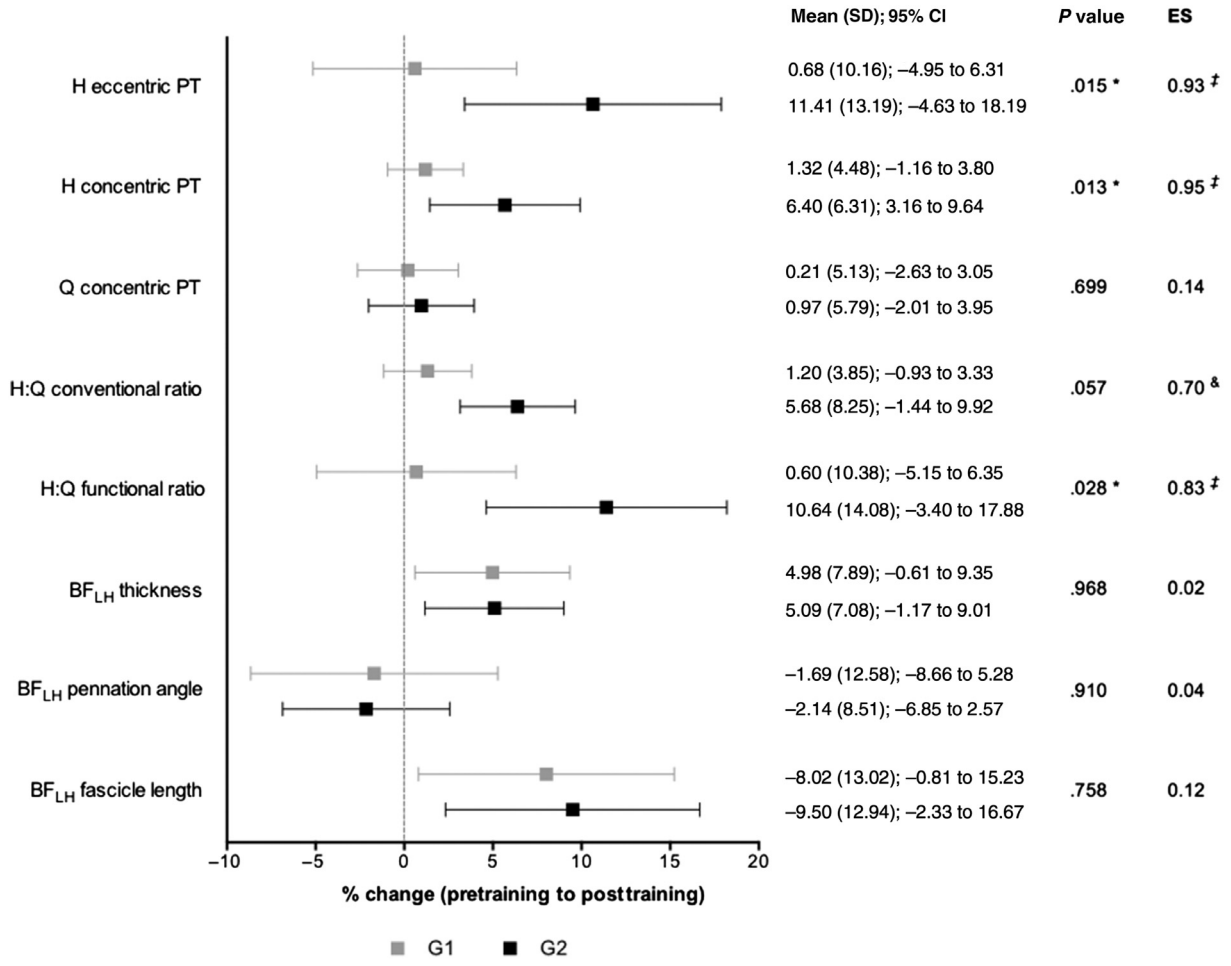

Figure 3 — Percentage changes (pretraining to posttraining) of muscle architecture and strength outcomes in G1 (NHE training once a week) and G2 (NHE training twice a week). BF_{LH} indicates biceps femoris long head; CI, confidence interval; ES, between-groups effect size; G1, group 1; G2, group 2; H, hamstring; PT, peak torque; Q, quadriceps; *Significant difference. &Moderate effect size. ‡Large effect size.

BF_{LH} fascicle length has been recently added to the list of potential risk factors for HSI. A retrospective investigation reported that athletes with a unilateral HSI history have shorter BF_{LH} fascicles in the previously injured limb than the contralateral uninjured limb,[40] and a prospective study demonstrated that professional football players with shorter BF_{LH} fascicles were 4 times more likely to sustain a future HSI than those with longer fascicles.[11] BF_{LH} ultrasonography appears as a relatively accessible and safe tool for clubs to track athletes at greater risk of HSI injury.[32]

Evidence supports that eccentric training induces addition of serially arranged sarcomeres (sarcomerogenesis) in animal models[41] and leads to increased fascicle length of hamstrings[42] and other muscle groups[43] in humans. NHE training is capable of increasing BF_{LH} fascicle length by 13% to 24% in recreationally active populations.[16,19,21] The high-level football players of both groups engaged in our study had fascicle length mean increases of ~8% to 9% (~1 cm). However, it is currently unclear if exercise induced increases in fascicle length are protective of HSI.

A recent systematic review concluded that reducing NHE volume prescription does not negatively affect adaptations in eccentric strength and muscle architecture when compared with high-dose interventions.[44] However, to the best of our knowledge, only 2 previous studies[29,31] compared muscular adaptations of NHE programs with different weekly training volumes. Unlike our findings, both studies suggest that a low volume is likely as effective as a high-volume training to improve hamstring muscle strength.[29,31] Interestingly, volunteers allocated to G2 in our study performed twice as many repetitions as those in G1 (~472 vs ~236 repetitions), whereas the groups with high training volume performed 3 to 4 times more repetitions than low-volume groups in studies by Presland et al[29] (440 vs 128 repetitions) and Lacome et al[31] (96 vs 24 repetitions). In other words, previous studies[29,31] applied NHE training programs with a greater volume difference and found similar strengthening levels between low-volume and high-volume groups, whereas our study found different responses on hamstrings strength with training programs with more similar volumes. These inconsistent findings highlight the need for further investigations as the level of competition and sport may have a differential effect on whether higher volume is more beneficial. On the other hand, our results further support the Presland's[29] and Lacome's et al[31] findings that BF_{LH} fascicle length can be increased with a low-volume NHE training program, suggesting the potential for HSI prevention even when applied at lower weekly volume.

The high number of players excluded due to injury (G1 = 8, G2 = 7) reduced the sample size in this study. However, this is expected when researchers develop a randomized trial into the

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real-world of competitive football. Despite evidence that performing NHE prior to football training may exacerbate HSI risk,[45] only 1 player (allocated in G1) sustained an HSI during the study period (~10 wk) and that injury did not occur during or after an NHE training session. Records of the club indicate a total of 7 HSIs in under-20 and under-23 teams in the first 10 weeks of the previous season when players were not engaged in a systematized training program with the NHE, suggesting a potential preventive effect of NHE on football players. It is also important to note that the 2 groups had different baseline strength values; up to 8% in some cases. Even though the statistical approach accounted for baseline values, it is still possible that the high-volume group had more space for improvement than the low-volume group.

Practical Applications

Given the association of both BF_{LH} fascicle length and hamstring strength with future risk of HSI, the current evidence highlights the importance of football players attending a minimum of 2-weekly NHE sessions to promote changes in both muscle structure and strength. However, a single weekly session of NHE may assist in HSI prevention through changes in muscle architecture when teams are unable to apply a greater volume of preventive training.

Conclusion

Our results support that only the NHE training program performed twice a week strengthened the hamstrings of high-level football players, while similar changes in muscle architecture occurred with both once- and twice-weekly sessions.

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